

5.4 Linear Functions of Random Variables

Example 15

See the data in Example 1.

$y \backslash x$	10	20	30	
10	0,05	0,05	0,20	0,30
20	0,10	0,20	0,10	0,40
30	0,10	0,15	0,05	0,30
	0,25	0,40	0,35	1

Let $Z = X + Y$ be the total score on language and numeracy. Determine $E(Z)$ and $V(Z)$.

Solution

- $$E(X + Y) = (10 + 10) \cdot 0.05 + (10 + 20) \cdot 0.05 + 40 \cdot 0.20 + 30 \cdot 0.10 + \dots + 60 \cdot 0.05$$

$$= 41$$
- $$V(X + Y) = E((X + Y)^2) - (E(X + Y))^2$$

$$= (10 + 10)^2 \cdot 0.05 + (10 + 20)^2 \cdot 0.05 + \dots + 60^2 \cdot 0.05 - 41^2$$

$$= 1760 - 1681$$

$$= 79$$

Remarks

- $X + Y$ is called a linear combination of X and Y .
- In general: a linear combination of $X_1, X_2, \dots, X_n$ is written as:

$$Y = c_1 \cdot X_1 + c_2 \cdot X_2 + \dots + c_n \cdot X_n$$

- For the data from Example 15 applies:

- $$E(X + Y) = E(X) + E(Y) \quad (41 = 20 + 21)$$
- $$V(X + Y) \neq V(X) + V(Y) \quad (119 \neq 60 + 59)$$

Theorem

Let X and Y be arbitrary random variables. Then:

$$E(X + Y) = E(X) + E(Y)$$

$$V(X + Y) = V(X) + V(Y) + 2 \cdot \text{cov}(X, Y)$$

In the special case of independency between X and Y :

$$V(X + Y) = V(X) + V(Y)$$

Proof

$$\begin{aligned}
\bullet \quad E(X + Y) &= \sum_x \sum_y (x + y) \cdot P(X = x, Y = y) \\
&= \sum_x \sum_y \{x \cdot P(X = x, Y = y) + y \cdot P(X = x, Y = y)\} \\
&= \sum_x \sum_y x \cdot P(X = x, Y = y) + \sum_x \sum_y y \cdot P(X = x, Y = y) \\
&= \sum_x x \cdot P(X = x) + \sum_y y \cdot P(Y = y) \\
&= E(X) + E(Y) \\
\bullet \quad V(X + Y) &= E((X + Y)^2) - (E(X + Y))^2 \\
&= E(X^2 + 2XY + Y^2) - \{E(X)^2 + 2 \cdot E(X) \cdot E(Y) + E(Y)^2\} \\
&= E(X^2) - E(X)^2 + E(Y^2) - E(Y)^2 + 2 \cdot E(XY) - 2 \cdot E(X) \cdot E(Y) \\
&= V(X) + V(Y) + 2 \cdot \text{cov}(X, Y)
\end{aligned}$$

Remark

In general:

$$\begin{aligned}
\bullet \quad E(c_1 \cdot X_1 + c_2 \cdot X_2 + \dots + c_n \cdot X_n) &= c_1 \cdot E(X_1) + c_2 \cdot E(X_2) + \dots + c_n \cdot E(X_n) \\
\bullet \quad V(c_1 \cdot X_1 + c_2 \cdot X_2 + \dots + c_n \cdot X_n) &= c_1^2 \cdot V(X_1) + \dots + c_n^2 \cdot V(X_n) + 2 \cdot \sum_{i < j} \sum_j c_i c_j \text{cov}(X_i, X_j)
\end{aligned}$$

Sample Mean

The random variables $X_1, X_2, \dots, X_n$ are a [random sample of size \$n\$](#) if the X_i 's are independent random variables and every X_i has the same probability distribution.

The random variable

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{1}{n} \sum_{i=1}^n X_i$$

is called the [sample mean](#).

Theorem

If $X_1, X_2, \dots, X_n$ is a random sample taken from a population with mean μ and variance σ^2 , then

$$\begin{aligned}
E(\bar{X}) &= \mu \\
V(\bar{X}) &= \sigma^2/n
\end{aligned}$$

Proof

$$\begin{aligned}
\bullet \quad E(\bar{X}) &= E\left(\frac{1}{n}X_1 + \frac{1}{n}X_2 + \dots + \frac{1}{n}X_n\right) \\
&= \frac{1}{n}E(X_1) + \frac{1}{n}E(X_2) + \dots + \frac{1}{n}E(X_n) \\
&= n \cdot \frac{1}{n}\mu = \mu \\
\bullet \quad V(\bar{X}) &= V\left(\frac{1}{n}X_1 + \frac{1}{n}X_2 + \dots + \frac{1}{n}X_n\right) \\
&= \frac{1}{n^2}V(X_1) + \frac{1}{n^2}V(X_2) + \dots + \frac{1}{n^2}V(X_n) + 2 \cdot \sum_{i < j} \sum_j \frac{1}{n^2} \text{cov}(X_i, X_j) \\
&= n \cdot \frac{1}{n^2}\sigma^2 + 2 \cdot 0 = \frac{\sigma^2}{n}
\end{aligned}$$

Using Linear Combinations in Case of Normal Distribution

Let $X_1, X_2, \dots, X_n$ be a random sample of a **normally distributed random variable X with parameters μ_X and σ_X** . Let Y be **normally distributed with parameters μ_Y and σ_Y** , and suppose that X and Y are **independent**.

Then

- $X_{\text{sum}} = X_1 + X_2 + \dots + X_n \sim N(\mu_{\text{sum}} = n \cdot \mu; \sigma_{\text{sum}}^2 = n \cdot \sigma^2)$
Therefore: $\sigma_{\text{sum}} = \sqrt{n} \cdot \sigma$
- $\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} \sim N(\mu_{\text{mean}} = \mu; \sigma_{\text{mean}}^2 = \frac{\sigma^2}{n})$
Therefore: $\sigma_{\text{mean}} = \frac{\sigma}{\sqrt{n}}$
- $Z = 2X_1 + 3X_2 \sim N(\mu_Z = 5\mu; \sigma_Z^2 = 13\sigma^2)$
- $D = X - Y \sim N(\mu_D = \mu_X - \mu_Y; \sigma_D^2 = \sigma_X^2 + \sigma_Y^2)$

Proof

Since $D = X - Y = 1 \cdot X + (-1) \cdot Y$, it follows

$$\begin{aligned} \mu_D &= 1 \cdot \mu_X + (-1) \cdot \mu_Y = \mu_X - \mu_Y \\ \sigma_D^2 &= 1^2 \cdot \sigma_X^2 + (-1)^2 \cdot \sigma_Y^2 = \sigma_X^2 + \sigma_Y^2 \end{aligned}$$

Example 16

Content X of a can of beer is **normally distributed** with parameters $\mu = 330$ ml and $\sigma = 1$ ml.

- Determine the probability that the average content of two cans is less than 329 ml.
- Determine the probability that the difference between two cans is more than 5 ml.

Solution

- $\bar{X} = \frac{X_1 + X_2}{2} \sim N(\mu = 330; \sigma = \frac{1}{\sqrt{2}} = 0.7071)$

So

$$\begin{aligned} P(\bar{X} < 329) &= \text{normalcdf}(-10^{10}, 329, 33, 0.7071) \\ &= 0.0786 \end{aligned}$$

- $D = X_1 - X_2 \sim N(\mu = 330 - 330 = 0; \sigma = \sqrt{2} \cdot 1 = 1.4142)$

So

$$\begin{aligned} P(D > 5) &= \text{normalcdf}(5, 10^{10}, 0, 1.4142) \\ &= 0.0002 \end{aligned}$$

$$\begin{aligned} \text{Requested probability} &= P(X_1 - X_2 > 5 \text{ of } X_2 - X_1 > 5) \\ &= 0.0004 \end{aligned}$$